NOTE: ALL CIVIL ENGINEERING CONSTRUCTION WORKS TO BE CARRIED OUT IN ACCORDANCE WITH AUTHORITY (INCLUDING LGA AND WATER AUTHORITY) STANDARDS AND GUIDELINES.THE AFOREMENTIONED GUIDELINES INCLUSIVE OF ALL SPECIFICATIONS TAKE PRECEDENCE OVER NOTES PROVIDED BELOW.

AUTHORITY APPROVALS AND DESIGN COORDINATION

1. REFER TO NORTHROP EROSION AND SEDIMENT CONTROL PLANS (C-120 AND 130 SERIES) FOR NOTES,LAYOUT & CONSTRUCTION DETAILS. IF THERE ARE QUESTIONS OR PROBLEMS WITH THE LOCATION, EXTENT, OR METHOD OF INSTALLATION, CONTACT THE ENGINEER OR RESPONSIBLE ON-SITE OFFICER FOR ASSISTANCE.
2. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO DEVELOP AND IMPLEMENT A SUITABLE EROSION AND SEDIMENT CONTROL PROGRAM IN ACCORDANCE WITH THE CURRENT LOCAL COUNCIL STANDARDS AND INTERNATIONAL EROSION CONTROL ASSOCIATION (IECA) GUIDELINES AND STANDARD DETAILS. EROSION AND SEDIMENT CONTROL DESIGNS MAY NEED TO BE AMENDED IN RESPONSE TO SITE CONDITIONS, IN WHICH CASE THE CONTRACTOR IS TO 'RED PEN MARK UP' THIS PLAN AND TO ENSURE THEY HAVE AN ADEQUATE/COMPLIANT SET OF EROSION AND SEDIMENT CONTROL PLANS ONSITE AT ALL TIMES DURING CONSTRUCTION.
3. EROSION & SEDIMENT CONTROL PROTECTION MEASURES SHALL BE MAINTAINED BY THE CONTRACTOR THROUGHOUT THE CONTRACT.
4. THE CONTRACTOR SHALL INFORM ALL SUB CONTRACTORS OF THEIR RESPONSIBILITIES IN MINIMISING THE POTENTIAL FOR SOIL EROSION AND POLLUTION TO DOWNSLOPE LANDS AND WATERWAYS.
5. UNDERTAKE SITE DEVELOPMENT WORKS SO THAT LAND DISTURBANCE IS CONFINED TO AREAS OF MINIMUM WORKABLE SIZE.

ACCESS AND SAFETY

1. THE CONTRACTOR SHALL COMPLY WITH ALL STATUTORY AND INDUSTRIAL REQUIREMENTS FOR PROVISION OF A SAFE WORKING ENVIRONMENT INCLUDING TRAFFIC CONTROL.
2. THE CONTRACTOR SHALL PROVIDE TRAFFIC MANAGEMENT PLANS FOR THE PROPOSED WORKS COMPLETED BY A SUITABLY QUALIFIED PERSON AND APPROVED BY COUNCIL / REGULATORY AUTHORITY. WORK IS NOT TO COMMENCE ON SITE PRIOR TO APPROVAL OF TRAFFIC MANAGEMENT SCHEME.
3. THE CONTRACTOR SHALL ENSURE THAT AT ALL TIMES ACCESS TO BUILDINGS ADJACENT THE WORKS IS NOT DISRUPTED.
4. WHERE NECESSARY THE CONTRACTOR SHALL PROVIDE SAFE PASSAGE OF VEHICLES AND/OR PEDESTRIANS THROUGH OR BY THE SITE.
5. THE CONTRACTOR SHALL ENSURE PUBLIC ACCESS EXTERNAL TO THE SITE IS IN ACCORDANCE WITH COUNCILS REQUIREMENTS.

CLEARING, ENVIRONMENT AND ECOLOGY

1. REFER TO LANDSCAPE / ARCHITECTS PLAN FOR TREES TO BE RETAINED AND PROTECTED.
2. ANY EXISTING TREES WHICH FORM PART OF THE FINAL LANDSCAPING PLAN SHALL BE PROTECTED FROM CONSTRUCTION ACTIVITIES BY;
   - 2.1. PROTECTING THEM WITH BARRIER FENCING OR SIMILAR MATERIALS INSTALLED OUTSIDE THE DRIP LINE.
   - 2.2. ENSURING THAT NOTHING IS NAILED TO ANY PART OF THE TREE.
   - 2.3. CARE IS TAKEN NOT TO CUT ROOTS UNNECESSARILY. COUNCILS AND/OR INDEPENDENT ARBORISTS TO BE CONSULTED WHERE TREE ROOTS ARE TO BE REMOVED AND/OR CUT.
3. PRIOR TO COMMENCEMENT OF WOKS CONTRACTOR IS TO:
   3(A). ENSURE ALL APPROVALS AND PERMITS IN PLACE
   3(B). PRESTART MEETINGS HAVE BEEN HELD
   3(C). SUPERINTENDENT, GEOTECHNICAL ENGINEER & NORTHROP ARE NOTIFIED.
   3(D). ALL EROSION &SEDIMENT CONTROL MEASURES ARE IN PLACE
   3(E). LOCATE ALL EXISTING SERVICES TO QL-A.
   3(F). REVIEW GEOTECHNICAL REPORT & (ALLOW FOR) GEOTECHNICAL ENGINEER METHODOLOGY FOR FILLING & EXCAVATION WORKS.
4. AN AUTHORISED SPOTTER-CATCHER IS TO BE ON SITE PRIOR TO THE COMMENCEMENT OF & DURING THE CLEARING WORK.
5. ANY CLEARING REQUIRED TO BE UNDERTAKEN BY THE CONTRACTOR IS TO BE STRICTLY IN ACCORDANCE WITH THE COUNCIL APPROVED VEGETATION MANAGEMENT PLAN.
6. THE AREA WITHIN THE LIMITS OF CLEARING SHALL BE CLEARED OF ALL VEGETATION, ALL MINOR MAN-MADE STRUCTURES (SUCH AS FENCES, BUILDING MATERIAL & EXISTING DRIVEWAYS), ALL RUBBISH & OTHER MATERIALS WHICH, IN THE OPINION OF THE SUPERINTENDENT, ARE UNSUITABLE FOR USE IN THE WORKS WITH THE EXCEPTION OF CERTAIN TREES MARKED FOR PRESERVATION.
7. TREES ARE TO BE MULCHED & STOCKPILED ON SITE FOR USE IN LANDSCAPING.
8. WHERE PARTS OF TREES CANNOT BE MULCHED, THEY ARE TO BE DISPOSED OF OFF-SITE AT AN APPROVED FACILITY.
9. ALL TOPSOIL FROM EARTHWORKS AREA SHALL BE STRIPPED & STOCKPILED PRIOR TO THE COMMENCEMENT OF ANY EARTHWORKS OPERATIONS.
10. NOTWITHSTANDING THE LIMITS OF CUTTING & FILLING SHOWN ON THE DRAWINGS, THE ACTUAL LIMITS SHALL BE DETERMINED ON SITE BY THE ENGINEER DURING CONSTRUCTION. FINISHED SURFACE CONTOURS MAY BE ADJUSTED BY WRITTEN DIRECTION OF THE ENGINEER DURING CONSTRUCTION.
11. ALL NEW WORKS TO MATCH NEATLY WITH EXISTING. THE CONTRACTOR SHALL CONFIRM LOCATION OF ALL EXISTING SERVICES & PROTECT THESE SERVICES DURING CONSTRUCTION. DAMAGED SERVICES SHALL BE REPAIRED AT THE CONTRACTORS EXPENSE.
12. ALL OF THE SITE SHALL BE KEPT FREE DRAINING DURING ALL PHASES OF THE PROJECT.
13. ALL DISTURBED AREAS TO BE GRASS SEEDED WITH MINIMUM 80% COVERAGE. SEEDING TO OCCUR WITHIN 7 DAYS OF COMPLETION OF EARTHWORKS.

EROSION AND SEDIMENT CONTROL

1. REFER TO NORTHROP EROSION AND SEDIMENT CONTROL PLANS (C-120 AND 130 SERIES) FOR NOTES,LAYOUT & CONSTRUCTION DETAILS. IF THERE ARE QUESTIONS OR PROBLEMS WITH THE LOCATION, EXTENT, OR METHOD OF INSTALLATION, CONTACT THE ENGINEER OR RESPONSIBLE ON-SITE OFFICER FOR ASSISTANCE.
2. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO DEVELOP AND IMPLEMENT A SUITABLE EROSION AND SEDIMENT CONTROL PROGRAM IN ACCORDANCE WITH THE CURRENT LOCAL COUNCIL STANDARDS AND INTERNATIONAL EROSION CONTROL ASSOCIATION (IECA) GUIDELINES AND STANDARD DETAILS. EROSION AND SEDIMENT CONTROL DESIGNS MAY NEED TO BE AMENDED IN RESPONSE TO SITE CONDITIONS, IN WHICH CASE THE CONTRACTOR IS TO 'RED PEN MARK UP' THIS PLAN AND TO ENSURE THEY HAVE AN ADEQUATE/COMPLIANT SET OF EROSION AND SEDIMENT CONTROL PLANS ONSITE AT ALL TIMES DURING CONSTRUCTION.
3. EROSION & SEDIMENT CONTROL PROTECTION MEASURES SHALL BE MAINTAINED BY THE CONTRACTOR THROUGHOUT THE CONTRACT.
4. THE CONTRACTOR SHALL INFORM ALL SUB CONTRACTORS OF THEIR RESPONSIBILITIES IN MINIMISING THE POTENTIAL FOR SOIL EROSION AND POLLUTION TO DOWNSLOPE LANDS AND WATERWAYS.
5. UNDERTAKE SITE DEVELOPMENT WORKS SO THAT LAND DISTURBANCE IS CONFINED TO AREAS OF MINIMUM WORKABLE SIZE.

EARTHWORKS

1. EARTHWORKS IS TO BE CARRIED OUT IN ACCORDANCE WITH AS3798-2007 "GUIDELINES ON EARTHWORKS FOR COMMERICAL AND RESIDENTIAL DEVELOPMENTS.
2. AT THE COMMENCEMENT OF FILLING OPERATIONS FOR BULK EARTHWORKS A GEOTECHNICAL ENGINEER IS TO VISIT THE SITE & CONFIRM THE SUITABILITY OF THE METHODOLOGY OF ACHIEVING THE REQUIRED COMPACTION REQUIREMENTS.
3. STRIP TOPSOIL, VEGETABLE MATTER AND RUBBLE TO EXPOSE NATURALLY OCCURRING MATERIAL AND STOCKPILE ON SITE AS DIRECTED BY THE SUPERINTENDENT.
4. WHERE FILLING IS REQUIRED TO ACHIEVE DESIGN SUBGRADE, PROOF ROLL EXPOSED NATURAL SURFACE WITH A MINIMUM OF TEN PASSES OF A VIBRATING ROLLER (MINIMUM STATIC WEIGHT OF 10 TONNES) IN THE PRESENCE OF THE SUPERINTENDENT.
5. THE CONTRACTOR IS TO ALLOW FOR A SUITABLY QUALIFIED GEOTECHNICAL ENGINEER TO PROVIDE ADVICE AND CERTIFICATION OF ANY WORKS ASSOCIATED WITH TREATING OR MANAGING UNSUITABLE GROUND CONDITIONS THROUGHOUT THE CONTRACT (e.g. STABILITY OF EXCAVATIONS, POOR SUBGRADE, etc).
6. ALL SOFT OR WET MATERIAL ENCOUNTERED ON SITE THAT ARE CLASSIFIED AS UNUSABLE IN THEIR CURRENT STATE SHALL BE EXCAVATED AND DRIED TO A LEVEL WHERE IT CAN BE USED FOR ITS INTEDED PURPOSE AND ACHEIVE THE COMPACTION RATES REQUIRED.
7. ALL MATERIAL THAT SATISFIES THE REQUIRMENTS OF UNSUITABLE MATERIALS (AS 3798) IS TO BE REMOVED FROM SITE AS DIRECTED BY THE SUPERINTENDENT AND REPLACED WITH APPROVED MATERIAL SATISFYING THE REQUIREMENTS BELOW.
8. PROVIDE CERTIFICATES VERIFYING THE QUALITY OF IMPORTED MATERIAL FOR THE SUPERINTENDENTS APPROVAL.
9. ALL FILL MATERIAL SHALL BE PLACED IN MAXIMUM 200mm THICK LAYERS (LOOSE) AND COMPACTED AT OPTIMUM MOISTURE CONTENT (+ OR - 2%) TO ACHIEVE A DRY DENSITY DETERMINED IN ACCORDANCE WITH AS1289.2.1.1, AS1289.5.7.1 AND AS1289.5.8.8 OF NOT LESS THAN THE FOLLOWING STANDARD MINIMUM DRY DENSITY;

LOCATION	COMPACTION REQUIREMENT
RESIDENTIAL LOTS	95% SMDD (CS)*
COMMERCIAL, FILLS TO SUPPORT MINOR LOADINGS	98% SMDD (CS)*
FILL TO SUPPORT PAVEMENTS	
(a) GENERAL FILL	95% SMDD (CS)*
(b) SUBGRADE TO A DEPTH OF 0.3M	98% SMDD (CS)*

* COHESIVE SOILS

10. TESTING OF THE SUBGRADE SHALL BE CARRIED OUT BY AN APPROVED N.A.T.A. REGISTERED LABORATORY AT THE CONTRACTORS EXPENSE.
11. ALLOW THE FOLLOWING COMPACTION TESTING BY N.A.T.A. REGISTERED LABORATORY FOR PLATFORMS AND FILL LAYERS IN ACCORDANCE WITH THE LATEST VERSION OF AS3798 - 1 TEST PER LAYER PER MATERIAL TYPE PER 2500M² OR 1 TEST PER 500M³ DISTRIBUTED REASONABLY EVENLY THROUGHOUT FULL DEPTH AND AREA OR 3 TESTS PER LOT (WHICHEVER REQUIRES THE MOST TESTS).
12. WHERE TEST RESULTS ARE BELOW THE SPECIFIED COMPACTION, RECOMPACT AND RETEST UNTIL SPECIFIED COMPACTION STANDARDS ARE ACHIEVED, OTHERWISE SUBGRADE REPLACEMENT IS REQUIRED IF COMPACTION STANDARDS ARE NOT ACHIEVED.
13. ALLOW FOR EXCAVATION IN ALL MATERIALS AS FOUND U.N.O. NO ADDITIONAL PAYMENTS WILL BE MADE FOR EXCAVATION IN WET OR HARD GROUND.
14. WHERE THERE IS INSUFFICIENT EXCAVATED MATERIAL SUITABLE FOR FILLING OR SUBGRADE REPLACEMENT, THE CONTRACTOR IS TO ALLOW TO IMPORT FILL. IMPORTED FILL SHALL COMPLY WITH THE FOLLOWING;
   - 14.1. BE OF VIRGIN EXCAVATED NATURAL MATERIAL OR
   - 14.2. CONTRACTOR TO PROVIDE EVIDENCE IMPORT IS SUITABLE USE
   - 14.3. PLASTICITY INDEX BETWEEN 2-15% AND CBR > 8
   - 14.4. FREE FROM ORGANIC AND PERISHABLE MATTER

   MAXIMUM SIZE 50MM, PASSING 75 MICRON SIEVE (<25%)
15. THE CONTRACTOR SHALL PROGRAM THE EARTHWORKS OPERATION SO THAT THE WORKING AREAS ARE ADEQUATELY DRAINED DURING THE PERIOD OF CONSTRUCTION. THE SURFACE SHALL BE GRADED AND SEALED OFF TO REMOVE DEPRESSIONS, ROLLERS MARKS AND SIMILAR WHICH WOULD ALLOW WATER TO POND AND PENETRATE THE UNDERLYING MATERIAL. ANY DAMAGE RESULTING FROM THE CONTRACTOR NOT OBSERVING THESE REQUIREMENTS SHALL BE RECTIFIED AT THEIR COST.
16. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO ENSURE AND MAINTAIN THE INTEGRITY OF ALL SERVICES, CONDUITS AND PIPES DURING CONSTRUCTION, SPECIFICALLY DURING THE BACKFILLING AND COMPACTION PROCEDURE. ANY AND ALL DAMAGE TO NEW OR EXISTING SERVICES AS A RESULT OF THESE WORKS SHALL BE REPAIRED BY THE CONTRACTOR AT NO ADDITIONAL COST.

DEEP EXCAVATIONS

17. PRIOR TO THE COMMENCEMENT OF EXCAVATION WORKS GREATER THAN 1.5M IN DEPTH, THE CONTRACTOR SHALL OBTAIN THE SERVICES OF A SUITABLY QUALIFIED GEOTECHNICAL ENGINEER TO DETERMINE THE STABILITY OF THE NATURAL MATERIAL AND BENCHING REQUIREMENTS.
18. THE CONTRACTOR MUST PROVIDE THE SUPERINTENDENT AND OR THE DESIGN ENGINEER WITH A COPY OF THE GEOTECHNICAL ENGINEERS REPORT.
19. THE CONTRACTOR IS TO PROVIDE SAFETY BARRIERS / FENCING IN ACCORDANCE WITH OH&S AND REGULATORY AUTHORITY REQUIREMENTS.

SERVICE TRENCHES

20. BACKFILL UNDER EXISTING PAVEMENTS

    SAWCUT EXISTING SURFACES PRIOR TO EXCAVATION. BACKFILL ALL TRENCHES UNDER EXISTING ROADS WITH BEDDING MATERIAL TO 300MM MINIMUM ABOVE SERVICE. PROVIDE APPROVED BACKFILL MATERIAL TO WITHIN 350MM MINIMUM OF FINISHED SURFACE, PROVIDE 300MM MINIMUM LEAN MIX CONCRETE TO UNDERSIDE OF ASPHALTIC CONCRETE.
21. BACKFILL UNDER PROPOSED PAVEMENTS

    PLACE BEDDING MATERIAL TO 300MM MINIMUM ABOVE SERVICE. PLACE APPROVED BACKFILL MATERIAL TO UNDERSIDE OF PROPOSED PAVEMENT AS SPECIFIED.
22. BACKFILL UNDER FOOTPATHS, ALLOTMENTS AND LANDSCAPE AREAS

    PLACE BEDDING MATERIAL TO 300MM MINIMUM ABOVE SERVICE. PLACE APPROVED SELECT MATERIAL FROM EXCAVATION OR IMPORT TO THE SURFACE.
23. BEDDING MATERIAL, APPROVED BACKFILL MATERIAL AND SELECT MATERIAL SHALL COMPLY WITH THE FOLLOWING:

    (a) BEDDING MATERIAL (FOUNDATION, HAUNCH AND OVERLAY)

    SHALL HAVE THE FOLLOWING GRADING OR COURSE, CLEAN, SHARP RIVER SAND APPROVED BY THE SUPERINTENDENT:

AS SIEVE SIZE	% PASSING BY MASS
19.0	100
2.36	40 - 100
0.425	15 - 70
0.075	3 - 30

BEDDING COMPACTION

NON-COHESIVE MATERIAL - DENSITY INDEX OF 70 MIN, REFER AS1289.5.5.1

SAND - COMPACT BY FLOODING USE OF VIBRATORS

(b) BACKFILL TO CONSIST OF APPORVED SUBGRADE REPLACEMENT MATERIAL CBR 15 MIN, PLACED IN LAYERS NOT EXCEEDING 250MM LOOSE IN DEPTH AND SHALL BE COMPACTED TO 95% MMDD MIN.

(c) SELECT MATERIAL FROM EXCAVATIONS SHALL BE PLACED IN LAYERS NOT EXCEEDING 250MM LOOSE IN DEPTH SHALL BE COMPACTED TO 95% SMDD MIN.

24. ALLOW THE FOLLOWING COMPACTION TESTING BY N.A.T.A. REGISTERED LABORATORY FOR TRENCHES IN ACCORDANCE WITH THE LATEST VERSION OF AS3798 - 1 TEST PER 2 LAYERS PER 40 LINEAR METRES.

RAINWATER REUSE

1. PROVIDE RAINWATER RE-USE SYSTEM TO SUPPLY WATER FOR IRRIGATION.
2. GUTTER GUARD TO BE INSTALLED ON ALL EAVES GUTTERS.
3. PRESSURE PUMP / TAP TO BE PROVIDED FOR THE REUSE OF CAPTURED TANK WATER.
4. A PERMANENT SIGN IS TO BE LOCATED IN THE VICINITY OF THE TANK STATING THE WATER IS "NON POTABLE WATER" WITH APPROPRIATE HAZARD IDENTIFICATION.
5. ALL RAINWATER SERVICES SHALL BE CLEARLY LABELED "NON POTABLE WATER" WITH APPROPRIATE HAZARD IDENTIFICATION.
6. PIPEWORK USED FOR RAINWATER SERVICES SHALL BE COLOURED LILAC IN ACCORDANCE WITH AS1345.
7. ALL VALVES AND APERTURES SHALL BE CLEARLY AND PERMANENTLY LABELED WITH SAFETY SIGNS TO COMPLY WITH AS1319.
8. AN AIR GAP OR RPZD TO ENSURE BACKFLOW PREVENTION (IF MAINS 'TOP UP' / BYPASS UTILISED)
9. RAINWATER TANK RETICULATION SYSTEM AND MAINS WATER BYPASS ARRANGEMENT TO BE INSTALLED IN ACCORDANCE WITH AS/NZS 3500.1.2-2003 AND THE QUEENSLAND PLUMBING AND WASTEWATER CODE.
10. A FIRST FLUSH FILTRATION DEVICE IS TO BYPASS THE FIRST 1mm OF RAINWATER.

EXISTING SERVICES

1. REFER TO THE CURRENT SITE SURVEY PLAN FOR REFERENCE PSM TO BE USED AS REFERENCE FOR ALL SETOUT INFORMATION DURING CONSTRUCTION. CONTRACTOR TO CONFIRM SURVEY SETOUT REFERENCE WITH REGISTERED SURVEYOR PRIOR TO CONSTRUCTION, ALL LEVELS TO AUSTRALIAN HEIGHT DATUM (AHD). GRID COORDINATE SYSTEM AS SPECIFIED BY SURVEYOR.
2. ALL UTILITY SERVICES INDICATED ON THE DRAWINGS ORIGINATE FROM SUPPLIED DATA OR DIAL BEFORE YOU DIG SEARCHES, THEREFORE THEIR ACCURACY AND COMPLETENESS IS NOT GUARANTEED. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO DETERMINE AND CONFIRM THE LOCATION AND LEVEL OF ALL EXISTING SERVICES PRIOR TO THE COMMENCEMENT OF ANY WORK. ANY DISCREPANCIES SHALL BE REPORTED TO THE SUPERINTENDENT. CLEARANCES SHALL BE OBTAINED FROM THE RELEVANT SERVICE AUTHORITY. NOTE SERVICE AUTHORITY REQUIREMENTS FOR LOCATING OF SERVICES PRIOR TO COMMENCEMENT OF WORKS. NORTHROP ACCEPTS NO RESPONSIBILITY FOR THE ACCURACY OF EXISTING UNDERGROUND SERVICES WHICH ARE PLOTTED FROM AUTHORITY RECORDS. DOCUMENTED DESIGNS MAY BE SUBJECTED TO ONGOING CHANGES UNTIL RECEIPT AND REVIEW OF MINIMUM QUALITY 'LEVEL A' EXISTING SERVICE LOCATION RESULTS ALONG FULL LENGTH OF PROPOSED MAIN ALIGNMENTS. NORTHROP WILL NOT BE HELD LIABLE FOR COST INCREASE OR DESIGN/DOCUMENTATION TIME EXTENSION RESULTING FROM NECESSARY DESIGN CHANGES TO ACHIEVE AUTHORITY CODE COMPLIANCE.
3. CARE TO BE TAKEN WHEN EXCAVATING NEAR EXISTING SERVICES. NO MECHANICAL EXCAVATIONS AREA TO BE UNDERTAKEN OVER COMMUNICATION, GAS OR ELECTRICAL SERVICES. HAND EXCAVATION ONLY IN THESE AREAS.
4. THE CONTRACTOR SHALL PROTECT AND MAINTAIN ALL EXISTING SERVICES THAT ARE TO BE RETAINED IN THE VICINITY OF THE PROPOSED WORKS. ANY AND ALL DAMAGE TO THESE SERVICES AS A RESULT OF THESE WORKS SHALL BE REPAIRED BY THE CONTRACTOR UNDER THE DIRECTION OF THE SUPERINTENDENT AT THE CONTRACTORS EXPENSE.
5. THE CONTRACTOR SHALL ALLOW IN THE PROGRAM FOR THE ADJUSTMENT (IF REQUIRED) OF EXISTING SERVICES IN AREAS AFFECTED BY WORKS.
6. THE CONTRACTOR SHALL ALLOW IN THE PROGRAM FOR THE CAPPING OFF, EXCAVATION AND REMOVAL (IF REQUIRED) OF EXISTING SERVICES IN AREAS AFFECTED BY WORKS COLLECTED UNLESS OTHERWISE ON THE DRAWINGS OR BY THE SUPERINTENDENT.
7. THE CONTRACTOR SHALL ENSURE THAT AT ALL TIMES SERVICES TO ALL BUILDINGS NOT AFFECTED BY THE WORKS ARE NOT DISRUPTED AND MAINTAINED.
8. PRIOR TO COMMENCEMENT OF ANY WORKS THE CONTRACTOR SHALL GAIN APPROVAL OF THE RELEVANT SERVICE AUTHORITY FOR THE RELOCATION AND/OR CONSTRUCTION OF TEMPORARY SERVICES AND FOR ANY ASSOCIATED INTERRUPTION OF SUPPLY.
9. THE CONTRACTOR SHALL CONSTRUCT TEMPORARY SERVICES TO MAINTAIN EXISTING SUPPLY TO BUILDINGS REMAINING IN OPERATION DURING WORKS TO THE SATISFACTION AND APPROVAL OF THE SUPERINTENDENT. ONCE DIVERSION IS COMPLETE AND COMMISSIONED THE CONTRACTOR SHALL REMOVE ALL SUCH TEMPORARY SERVICES AND MAKE GOOD TO THE SATISFACTION OF THE SUPERINTENDENT.

PRECAST STORMWATER PITS

1. THE USE OF PRE-CAST STORMWATER DRAINAGE PITS IS NOT ACCEPTED WITHOUT CONFIRMATION BETWEEN NORTHROP ENGINEERS AND THE CONTRACTOR REGARDING QUALITY CONTROL AND CERTIFICATION OF FINISHES.
2. REFER MANUFACTURERS SPECIFICATIONS FOR INSTALLATION GUIDELINES.
3. PRECAST PIT TO BE PLACED ON MINIMUM 150mm THICK CONCRETE PAD AND BED MINIMUM 50mm WHILST CONCRETE IS STILL PARTIALLY WET.
4. ENSURE PENETRATION IS CORED THROUGH PIT FACE TO ALLOW CONNECTION.
5. ENSURE A SMOOTH SEALED FINISH AT PIPE CONNECTIONS BY HAND APPLYING CONCRETE AROUND THE PIPE ON THE INTERNAL FACE OF THE PIT TO FILL IN ANY VOIDS CREATED WHEN PENETRATION FOR THE PIPE WAS CORED.
6. ENSURE A SEALED FINISH AT PIPE CONNECTIONS BY HAND-APPLYING MINIMUM 150mm THICK CONCRETE AROUND PIPE AT THE EXTERNAL FACE OF THE PIT. ENSURE CONCRETE DOES NOT AFFECT THE INTEGRITY OF THE SUBSOIL DRAINAGE CONNECTED TO THE PIT.
7. ENSURE PIPEWORK DOES NOT PROTRUDE INTO THE BEYOND THE WALL. PIPEWORK IS TO FINISH FLUSH WITH INTERNAL WALL (UNLESS OTHERWISE NOTED OR DETAILED).
8. ENSURE THE OUTLET PIPE IS CONNECTED AT THE INVERT LEVEL OF THE PIT TO DRAIN. ALTERNATIVELY FILL THE BASE OF THE PIT WITH MASS CONCRETE (MIN 50mm THICK) OR APPROVED GROUTING COMPOUND (LESS THAN 50mm THICK) TO DRAIN.
9. PROVIDE CONCRETE BENCHING TO SIDES OF PIT TO SUIT PIPE DIAMETER. HEIGHT TO MATCH MINIMUM 1/3 PIPE DIAMETER.

STORMWATER DRAINAGE

1. THE CLASS OF STORMWATER PIPES TO BE ADOPTED FOR CONSTRUCTION SHALL MATCH THE CONTRACTOR'S ADOPTED CONSTRUCTION METHOD.
2. PIPES SHALL CONFORM IN ALL RESPECTS TO AS 4058:2007 PRECAST CONCRETE PIPES.
3. PIPES UP TO 600mm DIAMETER SHALL HAVE SPIGOT AND SOCKET JOINTS WITH RUBBER RINGS. PIPES LARGER THAN 600mm DIAMETER SHALL BE FLUSH (REBATED) JOINTED UNLESS NOTED OTHERWISE.
4. ALL PIPES SHALL BE CLASS 2 RCP U.N.O. WHERE uPVC PIPES HAVE BEEN SPECIFIED, THE FOLLOWING CLASS PIPEWORK IS TO BE ADOPTED U.N.O. Ø100mm OR LESS TO BE CLASS 'SN10' AND ABOVE Ø100mm TO BE CLASS 'SN8'.
5. PIPES DAMAGED AS AN APPARENT RESULT OF HANDLING AND CRACKED IN MORE THAN ONE PLACE SO AS TO SHOW CLEARLY VISIBLE CRACKS (EXCEEDING 0.13mm INSIDE OR OUTSIDE) WILL BE REJECTED.
6. PIPES SHOWING CLEARLY VISIBLE SHRINKAGE CRACKS INSIDE OR OUTSIDE AND WHERE THE OPENING OF THE CRACK IS MORE THAN 0.127mm FOR A LENGTH OF 300mm OR MORE WILL BE REJECTED.
7. PIPES SHALL BE JOINTED IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
8. uPVC STORMWATER LINES PASSING UNDER FLOOR SLABS TO BE CONCRETE ENCASED.
9. FRC PIPES EQUAL TO THAT OF THE STEEL REINFORCED CONCRETE PIPE CLASS SPECIFIED ON THE DRAWINGS MAY BE USED SUBJECT TO APPROVAL FROM THE SUPERINTENDENT.
10. ALL PIPE ARE TO BE LAID AT 1.0% MIN GRADE U.N.O.
11. COVERS
    - 11.1. USE CAST IRON SOLID COVERS COMPLYING WITH RELEVANT COUNCIL AND AUSTRALIAN STANDARDS WHICH ARE TO BE THE BEST QUALITY CAST IRON, FREE FROM CRACKS, FLAWS AND POROUS SPOTS. THE INITIALS SW ARE TO BE CLEARLY VISIBLE.
    - 11.2. COVERS ARE TO BE CAST IRON WHERE ACCESS CHAMBERS SITUATED WITHIN THE ROAD BOUNDARIES OR OTHER TRAFFICABLE AREAS AND CONCRETE INFILLED ELSEWHERE.
    - 11.3. ALL COVERS AND GRATES TO BE POSITIONED IN A FRAME AND MANUFACTURED AS A UNIT.
    - 11.4. ALL COVERS TO BE FITTED WITH POSITIVE COVER LIFTING KEYS
    - 11.5. CONCRETE INFILLED COVERS TO CONSIST OF CROSS-WEBBED, CELLULAR CONSTRUCTION WITH THE RIBS UPPERMOST TO ALLOW INFILLING WITH CONCRETE. INSTALL POSITIVE COVER LIFTING KEYS AND PLASTIC PLUGS.
    - 11.6. UNLESS DETAILED OR SPECIFIED OTHERWISE, COVERS AND GRATES TO BE CLASS 'D' IN VEHICULAR PAVEMENTS AND CLASS 'B' ELSEWHERE.
    - 11.7. ALL GRATED TRENCH DRAINS SHALL BE 'CLASS D' CAST IRON WITHIN VEHICULAR PAVEMENTS AND CLASS 'B' HEEL SAFE WITHIN PEDESTRIAN PAVEMENTS.
    - 11.8. ALL COVERS AND GRATES SHALL FINISH FLUSH WITH THE ADJACENT FINISHED SURFACE LEVELS.
12. ALL PIPE BENDS, JUNCTIONS, ETC ARE TO BE PROVIDED USING PURPOSE MADE FITTINGS OR STORMWATER PITS.
13. ALL CONNECTIONS TO EXISTING DRAINAGE STRUCTURES SHALL BE MADE IN A TRADESMAN-LIKE MANNER AND CEMENT RENDERED TO ENSURE A SMOOTH FINISH.
14. STORMWATER PIPEWORK TO FINISH FLUSH WITH INTERNAL PIT WALLS AND MUST NOT PROTRUDE. CONNECTION TO BE NEATLY RENDERED.
15. THE CONTRACTOR SHALL SUPPLY AND INSTALL ALL FITTINGS AND SPECIALS INCLUDING VARIOUS PIPE ADAPTORS TO ENSURE PROPER CONNECTION BETWEEN DISSIMILAR PIPEWORK.
16. U.N.O. MATERIAL USED FOR BEDDING OF PIPES SHALL BE APPROVED NON-COHESIVE GRANULAR MATERIAL HAVING HIGH PERMEABILITY AND HIGH STABILITY WHEN SATURATED AND FREE OF ORGANIC AND CLAY MATERIAL.
17. WHERE TRENCHES ARE IN ROCK, THE PIPE SHALL BE BEDDED ON A MIN 50mm CONCRETE BED (OR 75mm THICK BED OF 12mm BLUE METAL) UNDER THE BARREL OF THE PIPE. THE PIPE COLLAR AT NO POINT SHALL BEAR ON THE ROCK.
18. BEDDING SHALL BE U.N.O TYPE HS2 UNDER ROADS AND H2 UNDER GENERAL AREAS IN ACCORDANCE WITH CURRENT RELEVANT INDUSTRY STANDARDS AND GUIDELINES.
19. THE CONTRACTOR SHALL ENSURE AND PROTECT THE INTEGRITY OF ALL STORMWATER PIPES DURING CONSTRUCTION. ANY AND ALL DAMAGE TO THESE PIPES AS A RESULT OF THESE WORKS SHALL BE REPAIRED BY THE CONTRACTOR UNDER THE DIRECTION OF THE SUPERINTENDENT AND AT NO EXTRA COST.
20. THE PIT GRATE LEVELS NOMINATED ARE TO THE LIP OF KERB. ACCESS CHAMBER COVERS LEVEL NOMINATED ARE TO CENTRE OF THE COVER. REFER DETAILS FOR CONFIRMATION.
21. INVERT LEVELS SHOWN ON THE DRAINAGE LONGITUDINAL SECTIONS ARE THE LEVELS CALCULATED AT THE CENTRE OF MANHOLES.

SUBSOIL DRAINAGE

22. Ø100mm SUBSOIL DRAINAGE LINES WITH NON-WOVEN GEOTEXTILE FILTER SOCK SURROUND SHALL BE CONNECTED TO A STORMWATER DRAINAGE PIT (AT MIN 1% LONGITUDINAL GRADE) AND PROVIDED IN THE FOLLOWING LOCATIONS;
    - 22.1. THE HIGH SIDE OF PROPOSED TRAFFICKED PAVEMENT AREAS.
    - 22.2. ALL PLANTER AND TREE BEDS PROPOSED ADJACENT TO PAVEMENT AREAS.
    - 22.3. BEHIND RETAINING WALLS IN ACCORDANCE WITH RETAINING WALL DETAILS.
    - 22.4. ALL OTHER AREAS SHOWN ON DRAWINGS.
    - 22.5. CONTRACTOR IS TO MAKE ALLOWANCE IN BOTH TENDER AND CONSTRUCTION COSTING TO ALLOW FOR SUBSURFACE DRAINAGE BEHIND ALL RETAINING WALLS / ABOVE LOCATIONS AND TO MAKE CONNECTION TO STORMWATER SYSTEM.
28. WHERE SUBSOIL DRAINAGE PASSES BENEATH BUILDINGS / PAVED AREAS AND/OR PAVEMENTS. CONTRACTOR TO ENSURE Ø100MM CLASS 'SN10' UPVC DRAINAGE LINE IS USED AND THAT PROPRIETARY FITTINGS ARE USED TO RECONNECT SUBSOIL DRAINAGE LINE.
29. THE CONTRACTOR SHALL INSTALL INSPECTION OPENINGS / CLEAROUTS TO ALL SUBSOIL DRAINAGE LINES AND DOWNPIPE LINES AS SPECIFIED ON DRAWINGS AND IN ACCORDANCE WITH COUNCIL SPECIFICATIONS AT MAXIMUM 30m CENTRES AND AT ALL UPSTREAM ENDPOINTS.
30. PROVIDE 3.0M LENGTH OF Ø100 SUBSOIL DRAINAGE LINE WRAPPED IN NON-WOVEN GEOTEXTILE FILTER FABRIC TO THE UPSTREAM SIDE OF STORMWATER PITS, LAID IN STORMWATER PIPE TRENCHES AND CONNECTED TO DRAINAGE PIT.
31. WHERE MIGRATORY SOILS ARE ENCOUNTERED WITHIN DRAINAGE TRENCHES, CONTRACTOR IS TO ALLOW FOR TRENCH EMBEDMENT MATERIAL TO BE FULLY WRAPPED WITH NON-WOVEN GEOTEXTILE FILTER FABRIC. LAY FABRIC AGAINST FLOOR AND WALLS SO THAT IT FULLY ENCASES THE EMBEDMENT. PROVIDE A MINIMUM OF 250mm OVERLAP AT ALL FABRIC JOINTS.
32. IN AREAS WHERE DUMPED / HAND PLACED ROCK IS USED AS A MEANS OF SCOUR PROTECTION, CONTRACTOR IS TO EXCAVATE A MINIMUM OF 100mm FROM PROPOSED SURFACE, LEVEL AND COMPACT SUBGRADE AS SPECIFIED. ROCK TO THEN BE PLACED ON GEOTEXTILE FILTER FABRIC.

NOT FOR CONSTRUCTION

NOT TO SCALE

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JD	PRELIMINARY DESIGN 30%	12.09.25	A
JD	ISSUED FOR TENDER	23.12.25	B
JD	ISSUED FOR TENDER	28.01.26	C
LQ	ISSUED FOR TENDER	30.01.26	D
MM	ISSUED FOR APPROVAL	11.06.26	E

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SKYGATE PLAY
BRISBANE AIRPORT

CIVIL SPECIFICATION NOTES
SHEET 1 OF 3

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